

Automated Legal Metrology Compliance & Enforcement System

SIH Problem Statement 34 — Project Script

Project Objective

A single digital platform connecting Citizens, Businesses, Inspectors, and the Controller, using OCR-based packaging scans and automated rule-based notice generation to make compliance preventive, consistent, and transparent.

1. The Problem

The Legal Metrology Act, 2011 protects consumers by making sure every packaged product declares things like MRP, net quantity, manufacturing/expiry date, and manufacturer details correctly. But in practice:

- Businesses rarely learn what's wrong with their packaging until an inspector catches them — by then, they've already broken the law and often didn't even know the rule existed.
- Getting proper guidance costs ₹20,000–₹30,000 in consultancy fees, which is completely out of reach for a small home-based business (like a Self-Help Group selling pickles or papad).
- Consumers, especially online shoppers, have no easy way to report packaging violations they notice.
- Offence records are scattered — the same brand can be caught for the same violation in two different cities and get treated as a “first offence” both times, since there's no shared record.
- This lack of transparency also creates room for corruption in how offences and penalties are recorded.

2. Our Solution, in One Line

A single digital platform connecting Citizens, Businesses, Inspectors, and the Controller (government) — using OCR-based packaging scans and automated, rule-based notice generation to make compliance preventive, consistent, and transparent instead of punitive and inconsistent.

3. Who Uses It, and How

A. Citizen (Consumer) — Web Dashboard

- Signs up with mobile OTP verification (keeps out fake/spam accounts).
- Files a complaint against a product or retailer: selects a clear violation category (Wrong MRP, Missing Net Quantity, No Manufacturing Date, etc.), uploads photos of the packaging and the invoice/bill, and enters the retailer's name and address.
- Can track their complaint step-by-step: Received → Assigned → Inspected → Notice Issued → Resolved.
- Receives an incentive only once the case is compounded or the business corrects the issue — this rewards genuine reporting and discourages false complaints.

- Over time, this also builds public awareness of the Act and gives the department a citizen-powered way to discover violators.

B. Business / Retailer — Mobile App

(Importers, exporters, home-based businesses, SHGs, small and large companies)

- Registers their business on the app.
- Can scan their own product packaging (2–3 angles, camera-based, just like the inspector's process) and get an instant self-compliance check — missing fields, wrong formats, anything that violates the Act, along with a suggested fix.
- This self-check is completely private. It's never visible to inspectors or the government — the whole point is to give a business a safe space to learn and fix issues on their own, without a consultant, before anyone ever inspects them.
- If a notice is issued against them, they can view it, submit supporting evidence, raise a dispute if they disagree, submit consent where required, re-upload corrected packaging photos to clear an Improvement Notice, and pay any penalty — all inside the same app.
- This is especially built for small businesses who've never had affordable access to this kind of legal guidance before.

C. Inspector — Mobile App (works offline)

- Selects the registered business being inspected and scans the product packaging from multiple angles.
- The system automatically extracts all required label information (MRP, net quantity, dates, manufacturer details, etc.) using OCR.
- The inspector reviews and corrects the extracted data before anything is finalized — this keeps every legal document built on accurate information, not just whatever the OCR happened to read.
- The system checks its database to see if this exact product or manufacturer has already been caught for a violation anywhere else in the country, and automatically applies First Offence or Second Offence rules.
- Notices are auto-drafted using NLP, pulling exact rule references from the legal knowledge base — the inspector reviews and signs digitally before issuing.

Notice Ladder

1. Improvement Notice — first flag; gives the business a chance to correct the issue.
 2. Seizure Notice — issued whenever evidence or product samples need to be seized (can happen at any stage).
 3. Panchanama — formal record of the violation with two independent witnesses, the applicable Act section, and the penalty amount.
 4. Compounded Order — final penalty order, sent to the Controller for approval before the business pays.
- The system tracks how many days are left before a case must escalate to the next stage, so nothing gets missed.
 - If a retailer names their supplier during an inspection, the system automatically opens a linked case against that supplier and routes it to the Controller to assign to the right jurisdiction's inspector — gradually building a supply-chain map of violations.

D. Controller (Government Officer) — Web Dashboard

- Sees the complete statewide picture: total complaints, first vs. second offence counts, region-wise case load, and compounded cases.
- Reviews every Compounded Order — can approve it or reject it with a comment, sending it back to the inspector for revision.
- Can mark a serious or repeat case for prosecution instead of compounding, when the Act doesn't allow compounding.
- Assigns supply-chain-linked inspection cases to the correct inspector.

4. The Legal Knowledge Base — What Makes All of This Possible

Underneath all four dashboards sits one critical piece: a structured digital version of the Legal Metrology Act, 2011 and the Legal Metrology (Packaged Commodities) Rules, 2011. Every mandatory label field, the exact Act/Rule section it comes from, and the penalty for violating it is stored here in a queryable format. This is what the OCR engine checks packaging against, what the NLP engine pulls from to write legally accurate notices, and what powers the whole first-offence/second-offence system. Without this, the OCR scan is just a text reader — this is what makes it a compliance engine.

5. Why This Reduces Corruption

Every action in the system — every notice issued, every offence recorded, every approval or rejection — is logged and visible to the Controller. Because offence history is centralized and automatic, an inspector can no longer quietly under-record a second offence as a first offence. At the same time, a genuine small business has a full digital record protecting it from being wrongly harassed.

6. Business Model

- Subscription for businesses, tiered by turnover — gives access to unlimited self-checks, digital record-keeping, and online case management.
- 1-2% commission from the government on the total penalty amount collected through the platform.

7. Technology Snapshot

Layer	Technology
Mobile Apps (Inspector, Business)	Flutter
Web Apps (Citizen, Controller)	Next.js
API Gateway & Core Backend	NestJS
Authentication	Keycloak (role-based access)
AI / OCR / NLP Engine	Python + FastAPI
Core Database	PostgreSQL + PostGIS (geo/jurisdiction data)

Layer	Technology
Supply Chain Mapping	Neo4j (graph database)
Search	Elasticsearch
File Storage (evidence, images, PDFs)	S3 / MinIO
Async Job Queue (OCR processing)	RabbitMQ
Digital Signatures	eMudhra
Payments	Razorpay
Notifications	WhatsApp + Email

Prototype Scope — This document reflects the current, prototype-scoped version of the project — every feature described above is meant to be demonstrable, not aspirational.